

PingPong Paddle Tracker

Design & Progress Log — built with Claude

App version **v1.1** · Firmware version **v1.0** · Session date 2026-08-15

A wireless IMU attached to a table-tennis paddle streams motion to a phone app that detects ball hits and measures paddle-face speed. Logging is now driven by hits: every detected hit is saved as its own log. This document captures what has been built, the key technical decisions, and — most importantly — the roadmap of future improvements, so work can resume from exactly where it left off.

0. How to resume

- Read Section 6 (Future roadmap) first — that is the live to-do list; the flagship next step is spin / brush estimation (6.1).
- Sections 3–4 describe current firmware and app capabilities; Section 5 lists the tuned constants and empirical findings you will need.
- Key source files: **PingPongTrackerFirmware/src/main.cpp** (firmware); **ping_pong_tracker_app/lib/main.dart** (app + UI), **lib/motion_estimator.dart** (orientation, velocity, $\omega \times r$ face speed), **lib/hit_detector.dart** (vibration hit detector).
- Reference data: **data_v1.0/{hit,spin,swing}** are current clean captures (one stroke each, still at both ends); **data_v0.0/** holds the older, messier sets.

1. Project overview & goal

A Seeed XIAO nRF52840 Sense board (LSM6DS3TR-C IMU) is mounted on a table-tennis paddle. It streams accelerometer and gyroscope data over BLE at a fixed 1660 Hz to a Flutter phone app. All detection and analysis run in the app; the firmware is a thin, high-rate streamer.

The original goal — compare the time of maximum paddle speed to the time of ball contact — is complete: they are effectively simultaneous (peak rotational speed coincides with the hit to within ~15 ms). The project is now ready for more advanced, table-tennis-specific analytics (Section 6).

2. Hardware & architecture

- **MCU/Radio:** Seeed XIAO nRF52840 Sense (Nordic nRF52840, SoftDevice S140), Adafruit Bluefruit BLE stack.
- **IMU:** LSM6DS3TR-C, internal I²C (Wire1). Accelerometer ± 16 g, gyroscope 2000 °/s, both sampled at the shared max ODR of 1660 Hz.
- **Link:** BLE Nordic UART service; 2 Mbps PHY, 247-byte MTU, 7.5 ms connection interval, data-length extension — all to sustain the high sample rate.
- **App:** Flutter (Android, tested on a Pixel), flutter_blue_plus for BLE.
- **Design split:** the board only streams; the app does windowing, logging, hit detection, orientation/speed estimation, storage, and visualization.

3. Firmware capabilities (v1.0)

File: **PingPongTrackerFirmware/src/main.cpp** (PlatformIO, env *xiaoblesense*; flash with `pio run -t upload`).

Streaming

- Advertises as **PaddleTrack**. On connect, negotiates 2 Mbps PHY, 247-byte MTU, data-length extension, and a 7.5 ms interval for throughput.
- Reads the IMU on each new accel sample (XLDA), bursting all 12 bytes (gyro + accel) in one I²C read at 400 kHz.
- Batches samples into notifications on the UART TX characteristic. Packet: byte 0 = sample count N, byte 1 = battery %, then $N \times 12$ bytes of int16 LE ax, ay, az, gx, gy, gz (raw counts). Up to 20 samples/packet; flushes when full or after a 15 ms latency cap.
- Battery percentage sampled every 2 s from the on-board divider.

Versioning

- Firmware version exposed as a READ BLE characteristic (UUID ...0004...). The app reads it on connect and shows “fw v1.0”. Bump the FW_VERSION define and reflash when firmware changes.

Note: the firmware has intentionally stayed a pure streamer — no on-device detection or logging — so all logic can iterate in the app.

4. App capabilities (v1.1)

Files: **lib/main.dart** (UI, BLE, logging, storage), **lib/motion_estimator.dart** (orientation + speed), **lib/hit_detector.dart** (hit detection). Tabs: Connection, Logs, Settings.

4.1 Connection & live display

- Scan/connect to PaddleTrack; decodes packets and applies sensitivities (accel 0.488 mg/count, gyro 70 mdps/count).
- Live readouts: streaming status, sample rate, accel (g), gyro (°/s), orientation (roll/pitch/tilt), and live Face speed ($\omega \times r$) in m/s.
- Horizontal battery gauge + percentage under the sample rate when connected; app/firmware versions top-right (app v1.1 / fw v1.0).

4.2 Calibration

- “Calibrate (hold still)” averages ~1.5 s at rest to set gyro bias and the gravity-aligned initial orientation. Used by the orientation filter; the $\omega \times r$ face speed does not require it.

4.3 Logging — two modes (Settings toggle)

- **Automatic (default) — one log per hit:** each detected ball hit is saved as its own log. A ring buffer supplies the “before” history and recording continues the same amount after the hit, so every log spans $\pm(\text{window})$ around the hit — a single Window slider (default 1 s → 2 s total). Further hits inside a window stay in that log and are marked, not split out. (This replaced the earlier speed/acceleration-triggered capture.)
- **Manual:** a Start/Stop button on the Connection tab records straight to a log, auto-stopping after a timeout (0.1–5.0 s).
- Both modes share the same capture, finalize, and numbering code.

4.4 Hit detection

- Runs live on every sample (both modes). A ball impact rings the wood at high frequency (~80–830 Hz) while swing/hand motion is below ~80 Hz — cleanly separable.
- Algorithm (causal, real-time): 2-pole high-pass at 120 Hz on $|\text{accel}| \rightarrow 6$ ms RMS envelope $\rightarrow$ threshold with hysteresis ($0.4 \times$) and a 50 ms refractory.

- Tuned threshold 0.5 g: zero false positives on no-hit swings, catches every real hit. Threshold is a Settings slider (0.1–1.5 g).
- In automatic mode a detected hit is also the capture trigger (Section 4.3). Detected hit times attach to each log and are drawn as magenta vertical lines on all three graphs.

4.5 Paddle-face speed ($\omega \times r$, drift-free)

- Face speed = $|\omega \times r|$ — gyro (a direct measurement) crossed with a constant board-frame lever-arm vector r . No integration, so no drift; reads ~ 0 at rest by construction.
- r points along the handle axis, confirmed **board Z** by a pure handle-twist capture (gz dominant). So $\omega \times r$ excludes the handle-twist (spin) and keeps the swing $\rightarrow$ linear face-center speed.
- Lever-arm magnitude is a Settings slider (2–30 cm, default 12); sets only the m/s scale, not timing.
- Replaced the old accel-integrated speed (which drifted). The Speed chart overlays face $\omega \times r$ (indigo) vs accel (old, grey) as a built-in drift comparison.

4.6 Logs — charts, markers, management

- Each log stores time + 6 axes (columnar), a name, and hit times, as a binary file in app storage; survives restart. Numbering persists and restarts at 1 only when empty.
- Detail view: three charts — Speed (face $\omega \times r$ + old accel overlay), Accelerometer, Gyroscope — each with magenta hit markers, plus raw CSV rows.
- Rename (with automatic “(1)/(2)” de-dup), Share single (CSV), Share all (zip), Delete, Delete-all (with confirmation).
- “Always show logs list” toggle: returning to the Logs tab reopens the list, not the last-viewed log.

4.7 Settings (all persisted)

- Automatic-logging toggle; Hit threshold (g); Lever arm (cm); Auto-capture window ($\pm$ around each hit, s) or Manual auto-stop timeout (s); “Always show logs list”.

5. Key findings & reference values

Item	Value / finding
Sample rate (ODR)	1660 Hz (fixed)
Accel / gyro scale	0.488 mg/count (± 16 g) · 70 mdps/count (2000 °/s)
Logging trigger	each detected ball hit $\rightarrow$ its own log, $\pm$ window (default 1 s)
Hit detector	2-pole HP @ 120 Hz + 6 ms env; threshold 0.5 g; 50 ms refractory
Hit vs swing bands	impact ~ 80 –830 Hz; motion below 80 Hz ($\sim 5\times$ separation margin)
Handle axis	board Z (from pure-twist test: gz dominant ~ 528 °/s rms)
Face speed	$v = \omega \times r $, r along Z; drift-free; ~ 0 at rest
Lever arm default	12 cm (sets m/s scale only; timing scale-independent)
Peak speed vs hit	coincident within ~ 0 –40 ms across all v1.0 strokes
Old accel speed	drifts: peak often in follow-through, end 0.5–3.7 m/s even with pauses

Lever-arm calibration (how to set r): (a) tape-measure from the IMU to the paddle-face center along the handle (~ 12 –16 cm) for a good first value; (b) for a true m/s, film swings at 120–240 fps with a

ruler in frame, measure peak face speed, and set $r = v_{\text{peak}} / |\omega_{\text{peak}}|$ (ω in rad/s); (c) timing and relative comparisons do not depend on r . The $\omega \times r$ approximation is most accurate with the board mounted near the wrist/pivot.

6. Future roadmap — ways to keep improving (most important)

The peak-speed-vs-hit goal is done. Below are all proposed next directions, roughly ordered by value. Spin estimation (6.1) is the recommended flagship; impact mining (6.2) is a strong, mostly independent second track that finally puts the accelerometer to real use now that speed no longer relies on it.

6.1 Spin / brush estimation (flagship)

- Table tennis is about spin, and the most spin-relevant data is currently unused: (a) the handle-twist component (ω about board Z) excluded from linear speed is itself spin-generating; (b) the split of face velocity into normal vs tangential parts at contact indicates pace vs spin.
- Decompose v_{face} (from $\omega \times r$) into a normal component (into the ball $\rightarrow$ pace/drive) and a tangential component (brushing $\rightarrow$ spin). A high tangential/normal ratio is a “spin index”; a flat drive is mostly normal, a topspin loop mostly tangential.
- Feasible with what we have: the handle axis (Z) is known, so a single flat-on-table calibrate (gravity $\rightarrow$ face normal) yields the full paddle frame and the decomposition falls out.
- Deliverable: a per-stroke pace-vs-spin readout / spin index — makes this a genuine table-tennis tool rather than a generic motion logger.

6.2 Mine the accelerometer at impact

- **Hit hardness / impulse:** peak accel and the integral of the impact spike scale with how hard you hit — a “power” metric, and a rough outgoing-ball-speed estimate via momentum.
- **Sweet-spot / mishit detection:** an off-center hit induces a torque (gyro spike coincident with the accel spike) and a different vibration signature. The rotational-vs-linear response ratio gives a “clean contact” quality score.
- **Face angle at contact:** the orientation filter already knows the paddle’s tilt at the hit instant (open/closed face, angle of attack) — a key coaching metric, essentially free.
- **Vibration spectrum:** an FFT of the ~ 30 ms ring could separate clean vs edge hits, and might fingerprint different rubbers/blades.

6.3 Stroke classification

- Each auto-captured log is a clean single-stroke window — ideal labeled training data.
- From features (peak speed, peak ω , rotation axis, orientation change, pace/spin split, duration) build a rule-based or small ML classifier that auto-tags each log: forehand/backhand, drive, push, topspin loop, smash, serve.

6.4 Per-stroke & session analytics

- **Stroke report card:** speed, spin index, face angle, contact quality, contact timing, and stroke type — one summary per log.
- **Session dashboard:** stroke count, speed/spin distributions, stroke-type breakdown, and consistency (variance of peak speed and of contact timing). Consistency is often more coachable than peak numbers.
- **Contact-timing metric:** contact $\approx$ peak speed; the per-stroke deviation (hitting before your fastest = decelerating into the ball) is a useful drill metric.

6.5 Real-time feedback & drills

- After each captured hit, flash a “speed-gun” result (e.g. “Forehand · 6.2 m/s · topspin”).
- Target-based drills (“10 forehands over 5 m/s”), consistency challenges, with audio/haptic cues — turns the analysis tool into a practice aid.

6.6 Infrastructure, robustness & exports

- **Weak/spin hit robustness:** add a complementary cue — a sudden gyro impulse (torque from a glancing/brushy contact) — to catch weak high-spin hits that may fall below the 0.5 g vibration threshold (a known deferred concern; matters more now that hits trigger logging).
- **Full-frame calibration:** a guided flat-on-table (and optional on-edge) calibration to lock the paddle frame for the spin decomposition.
- **Export enrichment:** include computed metrics (peak face speed, hit time, spin index, face angle) in the CSV/zip exports, not just raw IMU.
- **On-device model:** as labeled captures accumulate, train a small classifier for stroke type / spin / quality.

7. Open items / where we left off

- Logging is now hit-triggered (one log per hit, ± 1 s default). Next feature — recommended: spin / brush estimation (6.1).
- Set the lever arm from a tape measure now; optionally do a one-time slow-mo video calibration for a true m/s (Section 5).
- Deferred: weak high-spin hits may slip under the 0.5 g hit threshold — now also affects whether a log is captured (6.6).
- The tangential/normal spin split needs one flat-on-table calibrate for the face normal (handle axis Z is already known).
- A 3D-printed handle mount is planned (rigid coupling to preserve hit vibration; re-tune the hit threshold with the case on if amplitude drops).

Generated as a design/handoff log so work can resume directly. Update the versions and roadmap as features land.